

- **What are the differences between Native, Hybrid, and Cross-platform mobile application development, and when is each approach commonly used?**

Mobile apps are an important part of everyday life because people use them for work, school, business, and entertainment. Because of this, choosing the right way to build a mobile app is an important step in any project. There are three common types of mobile app development, Native, Hybrid, and Cross-Platform. Each one has its own good points and limits. Knowing how they are different can help developers choose the best option based on the needs, budget, and goals of the project

- **What is the Flutter Ecosystem, and how does it help developers create modern mobile applications?**

Flutter is a open-source tool for creating mobile applications. It allows developers to build apps for different platforms using only one codebase. With Flutter the same app can work on Android, iOS, the web, and desktop without needing to write separate code for each platform

- **What is the Software Development Life Cycle (SDLC), and why is it important in building successful applications?**

The Software Development Life Cycle (SDLC) is a step-by-step process used to create software. It includes planning, designing, building, testing, launching, and taking care of the software after it released. This process helps developers stay organized and make sure the software meets the needs of users and the goals of the project

- **How do Flutter development practices and the SDLC work together in creating real-world mobile applications?**

In real mobile app development Flutter makes the Software Development Life Cycle (SDLC) easier because developers can use one codebase for different platforms. Instead of creating separate apps for Android and iOS, they only need to write the code once. This makes the development process simpler, saves time, and makes it easier to test, update, and release the application

- **Why should developers understand these concepts before starting an application project?**

Before starting a software project developers need to understand the basic ideas of software planning and design. This helps them make better decisions during development and avoid problems later good planning also makes the system easier to improve, supports the goals of the project, and reduces the need to fix major issues after the software is finished